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VoteKit: A Python package for computational social choice research

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Summary

The scholarly study of elections, known as *social choice theory*, centers on the provable properties of voting rules. Practical work in democracy reform focuses on designing or selecting systems of election to produce electoral outcomes that promote legitimacy and broad-based representation. For instance, the dominant electoral system in the United States is a one-person-one-vote/winner-take-all system, sometimes known as PSMD (plurality in single member districts); today, there is considerable reform momentum in favor of ranked choice voting because it is thought to mitigate the effects of vote-splitting and to strengthen prospects for minority representation, among other claimed properties.[1] Across the world, systems of election—and prospects for system change—vary substantially. From both a scholarly and a practical perspective, many questions arise about comparing the properties and tendencies of diverse systems of election in a rigorous manner.

`VoteKit` https://github.com/mggg/VoteKit is a Python package designed to facilitate just that kind of analysis, bringing together multiple types of functionality. Users can:

1. Create synthetic *preference profiles* (collections of ballots) with a choice of generative models and behavioral parameters;
2. Read in real-world *cast vote records* (CVRs) as observed examples of preference profiles; clean and process ballots, including by deduplication and handling of undervotes and overvotes;
3. Run a variety of *voting rules* to ingest preference profiles and output winner sets and rankings; and
4. Produce a wide range of *summary statistics* and *data visualizations* to compare and analyze profiles and election outcomes.

A tutorial that includes step-by-step example code can be found in the VoteKit documentation (MGGG Redistricting Lab, 2024b).

Statement of need

Since the 1990s, a fusion of economics and computer science has emerged under the name of *computational social choice*, studying questions of complexity and design and further advancing the axiomatic study of elections. But most of these innovations have been highly abstract, and there has been a significant gap in the literature—and in the landscape of software—between

[1] Recent ranked-choice voting reforms include the adoption of instant runoff voting (IRV) in Maine, Alaska, New York City, and single transferable vote (STV) in Portland, Oregon. Advocacy groups claiming various pro-democratic properties of ranked choice include Campaign Legal Center, FairVote, and many others.

the theory and the practice of democracy. On the software side, researchers have built a multitude of different packages for generating and analyzing elections.[2] Most packages, to our knowledge, handle just one part of the research arc; for instance, `PrefSampling` (Boehmer et al., 2024) generates profiles but does not conduct elections, while `VoteLib` (Šimbera, 2020) *only* conducts elections. Others, like `PrefLibTools` (Mattei & Rey, 2022) and `PrefVoting` (Holliday & Pacuit, 2025), provide support for generating profiles and conducting single-winner elections. Packages with multi-winner capability, like `abcvoting` (Lackner et al., 2023) or `Apportionment` (Lackner, 2022), do not support ranked voting. Note that single transferable voting (STV), a voting system actually used for political election in six countries, is curiously absent. `VoteKit` is built to provide an end-to-end pipeline that supports ranked, scored, and approval profiles as well as single- and multi-winner elections, with an emphasis on practical applicability.

Area of need: Generative models

For one concrete example of a literature and software gap, consider the construction of *generative models*. This term is often associated with large language models as paradigms of artificial intelligence; here, what is being generated is realistic voting rather than realistic language. In this setting, a generative model of voting is a probability distribution on the set of all possible ballots that can be cast in a given election style; profiles can be sampled from a generative model to produce simulated or synthetic elections. Having sources of rich, varied, and realistic data is essential to an empirically grounded research program to probe the properties of voting rules. Good generative models are also a fundamental tool to advise reformers deciding between alternative electoral systems in a new locality, as they enable generation of synthetic profiles keyed to the scale, demographics, and election styles considered for that specific place.

`VoteKit` implements many of the models typically used in computational social choice research,[3] as well as newer parametrized models that give users the ability to generate profiles that are designed to comport with real-world ranking behavior and particularly to generate polarized elections. Two leading choices are based on classic statistical ranking mechanisms, called the Plackett–Luce (PL) and Bradley–Terry (BT) models; another model called the Cambridge Sampler (CS) draws from historical ranking data in Cambridge, MA city council elections (Benadè et al., 2024). These models have flexible parameters—allowing users to vary voting bloc proportions, candidate strength within slates, and polarization between blocs—that can be specified or randomly sampled.

Area of need: Comparison and communication

Community groups looking to build local support for a shift in electoral systems often ask researchers to provide modeling studies that can help decide on a course of action—for example, when Portland, Oregon recently shifted its city council system to STV. `VoteKit` implements voting rules that stakeholders often seek to compare, with parameters designed to be tailored by the user to the specific locality. Available voting rules include:

- **Ranking-based (ordinal).** Plurality/SNTV, STV and IRV, (generalized) Borda, Alaska,[4]

[2] See for instance the extensive array of open-source tools on the Computational Social Choice (COMSOC) community page (Endriss & Rey, 2024) including the widely used collection of ranked data called PrefLib (Mattei & Rey, 2022). See also the materials provided by FairVote, including their DataVerse and GitHub (FairVote, 2024). The survey (Boehmer et al., 2024) provides an impressively comprehensive list of numerical experiments on elections. The PRAGMA Project (https://perma.cc/2P6V-8ZER) echoes our statement of need, noting that the current literature and software falls short in practical applicability and that the understanding of real and synthetic data is "very limited."

[3] Frequently used models include Impartial Culture (IC), Impartial Anonymous Culture (IAC), and spatial models. In a meta-analysis of 163 papers (Boehmer et al., 2024), the authors report that IC and Euclidean (spatial) models make up more than 75% of the election experiments found in 163 papers.

[4] Our implementation of the Alaska method is an SNTV/STV hybrid that uses single non-transferable vote to choose a set of finalists, then runs STV on the same preference profile to fill the seats. Alaska's elections run

Top-Two, Dominating sets/Smith method, Condo-Borda,[5] Sequential RCV.

- **Score-based (cardinal).** Range voting, Cumulative, Limited.
- **Approval-based (set).** Approval voting, Bloc plurality.

See generally (Amorós et al., 2016; Emerson, 2013; McCune et al., 2023; Reynolds et al., 2008; Tideman, 1995) for references. This list does not include every method that has attracted theoretical investigation; rather, it is oriented to methods used or considered for political representation, such as the final-four system in Alaska or the sequential RCV in Utah local elections. In addition, `VoteKit` is flexible enough to allow users to write custom voting rules.

Reform advocates also need to describe voting mechanisms and their likely outcomes effectively to members of their communities. The end-to-end pipeline provided by `VoteKit` allows advocates to toggle different system settings and compare expected outcomes. For example, Figure 1 is reprinted with permission from a report on reform proposals for the chambers of the Washington state legislature. Using the codebase that formed the foundation of `VoteKit`, researchers compared the expected outcomes for minority representation under six possible electoral systems.

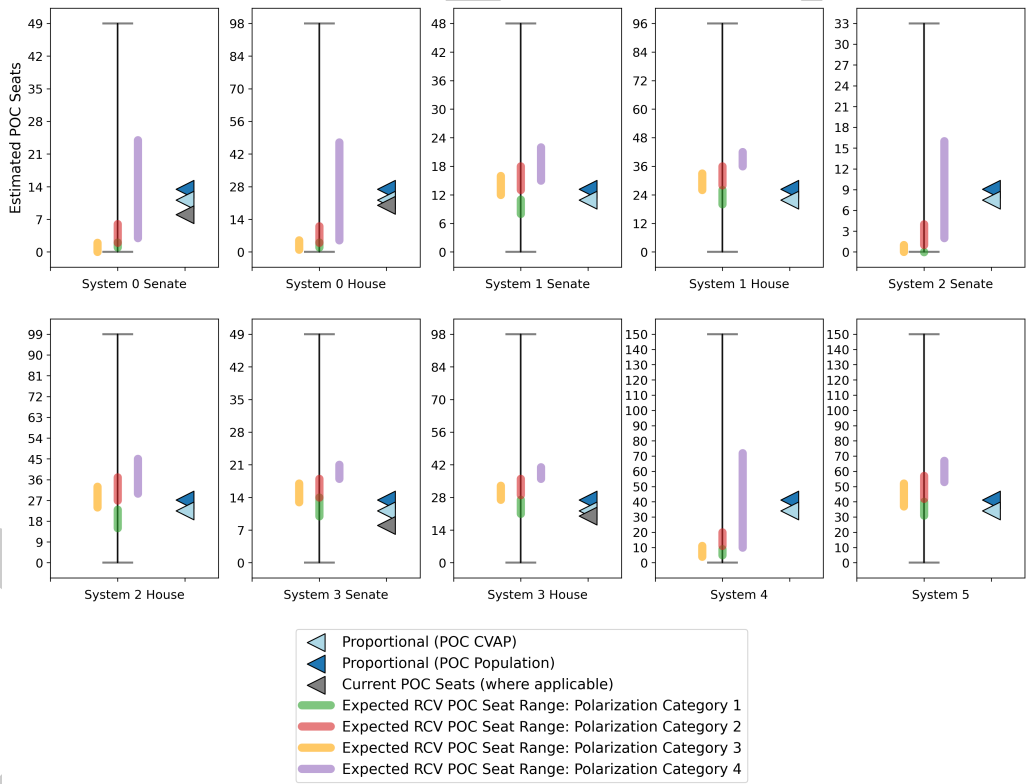

Figure 1: A comparison of a variety of electoral systems and their effect on minority representation, reprinted with permission, from a case study of reform proposals for the Washington state legislature (MGGG Redistricting Lab, 2021d). Even within ranked-choice proposals, certain options, like System 0 (based on single-member districts), are projected to be less successful for minority representation, while other systems, like System 1 (based on multi-member districts), predict that candidates of choice for people of color ("POC") are elected more in line with the POC share of population or citizen voting age population ("CVAP").

this in two distinct stages with four finalists and one seat; the top-two system amounts to running this with two finalists and one seat.

[5]Here, candidates are ordered by dominating sets (so that earlier ones in the list beat later ones in the list head-to-head), and ties are broken by Borda score. Note that this is distinct from Black's method (Black, 1986), which uses Borda score as a backup system in case the smallest dominating set is not a singleton.

Area of need: Resources for research

Previous research works such as Elkind et al. (2017) have compared properties of generative models; `VoteKit` has functionality to fully replicate this work and facilitates robust comparisons across a more comprehensive and up-to-date list of models. It also offers new analytical tools that will support research on elections. Some examples are shown in Figure 2. At left is a *ballot graph*, which shows the possible ballots, connected by elementary moves. At right is a visualization of similarity and difference between profiles produced by various generative methods, enabling comparisons in the style of Szufa et al. (2020).

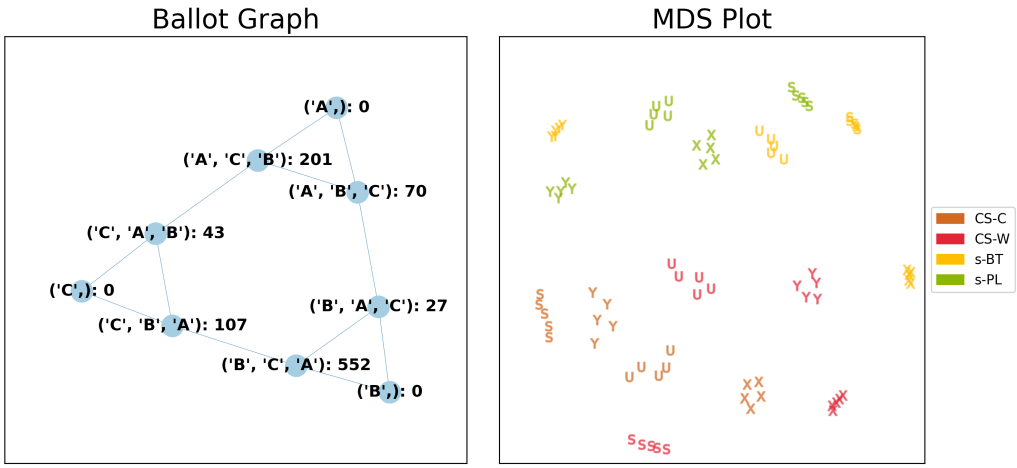

Figure 2: At left, the ballot graph for a 3-candidate election. The edges record swap moves and extension/truncation. (Note that the ballot $A > B > C$ is identified with the ballot $A > B$, since they are informationally equivalent in putting C last.) At right, a multidimensional scaling (MDS) plot shows similarity and difference among 80 synthetic profiles of 1000 ballots each, made with variations on the Cambridge Sampler (CS), Bradley-Terry (BT), and Plackett-Luce (PL) models. Compare Figure 1 in (Benadè et al., 2024). The letters represent different parameter settings related to candidate strength, and the image shows that these parameters create substantially different profiles, measured by L^1 difference in distributions.

Finally, `VoteKit` interacts seamlessly with a wide range of actual vote data, such as thousands of political elections collected by FairVote and a cleaned repository of over 1000 Scottish STV local government elections (FairVote, 2024; MGGG Redistricting Lab, 2023). Previously, the use of real data in election research was often extremely limited; for instance, a recent survey reports that the single most popular "real-life" dataset has been a survey of 5000 respondents' sushi preferences (Boehmer et al., 2024).

Projects

A significant number of white papers and scholarly articles by members of the MGGG Redistricting Lab and collaborators have used `VoteKit` (and its predecessor codebase) in recent years. These include the following.

- A large number of case studies in ranked-choice modeling, such as studies for the city councils of Chicago, IL (MGGG Redistricting Lab, 2019b) and Lowell, MA (MGGG Redistricting Lab, 2019a) and a range of jurisdictions across the Pacific Northwest (MGGG Redistricting Lab, 2021a, 2021d, 2021c, 2021b);
- A study modeling the impact of proposed legislation called the Fair Representation Act, which would convert U.S. Congressional elections to the single transferable vote system (MGGG Redistricting Lab, 2022);

- A detailed study isolating the impacts of varying hypotheses about voter behavior and candidate availability on the Massachusetts legislature (MGGG Redistricting Lab, 2024a);
- A peer-reviewed article for an election law audience on the impact of STV elections on minority representation (Benadè et al., 2021);
- A peer-reviewed article for a computer science and econ audience that probes whether STV delivers proportional representation (Benadè et al., 2024); and
- A peer-reviewed article for a computer science and operations research audience on optimizing to "learn" blocs and slates in real-world elections (Duchin et al., 2024).

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